

Hidden Markov Models

Machine Learning Techniques II

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Formative 2

1. Background and Motivation

Human Activity Recognition (HAR) automatically detects physical activities from the data of the smartphone sensor, enabling continuous health monitoring without the need of the user to do anything.

This project is an application of HAR in the monitoring of physiotherapy rehabilitation. Injured patients participate in programs to facilitate recovery, which involve rest, standing, walking and jumping exercises. But between clinic visits the extent of compliance is harder to monitor objectively, as the physiotherapists rely on self reported diaries. A HAR system based upon a smartphone would provide objective activity data on a daily basis, allowing for improvements in program modifications and earlier intervention.

Because human beings are engaged in sequences of activities over time with natural transitions between activities, a Hidden Markov Model (HMM) is used. Thus, the probabilistic framework is appropriate, and HMMs are suitable. The activity classes are the four types of activities that constitute a rehabilitation program (still, standing, walking and jumping).

2. Data Collection and Preprocessing

2.1 Data Collection

The data was gathered via Samsung SM-S901U with the recommended application Sensor Logger at data rate of 100 Hz from accelerometer (m/s^2) and gyroscope (rad/s) readings from all three axes simultaneously. A total of 4 activities were recorded for each 5-10 seconds, making a total of 51 recordings in the indoor and outdoor environments. The phone's position in the different activities was consistent with usage.

Each recording was saved as a zip archive made up of separate sensor CSVs. The loading function took both files and renamed the columns obviously, for two 100 Hz running sensors, it aligned them by row index.

2.2 Windowing

A window size of 100 samples (1 second at 100Hz) would contain at least one gait cycle but not be so long as to miss any gait cycle transitions. After increasing the training windows by a factor of 2, the 50-sample step (50% overlap) yielded 889 windows: 237 of still, 196 of standing, 253 of walking and 203 of jumping.

2.3 Feature Extraction

A 35-Dimensional feature vector was calculated for each window. The orientation, the variability, the energy and the intensity of the movement were measured in the time domain by the use of mean, standard deviation, mean of the square root and Signal Magnitude Area (SMA) per axis. Inter-axis coordination patterns were obtained from pairwise accelerometer axis correlations. The FFT was used to extract frequency domain features such as dominant frequency, spectral energy, and band energies of 0-3 Hz and 3-10 Hz which represent locomotion rhythms and impact transients.

2.4 Normalisation

All features were standardized by Z-Score, which scaled all the features to the mean of zero and a variance of one, to reduce the domination of high-magnitude features on HMM emission calculations. Scaler was fit on a training set and applied to a test set to avoid the leakage.

3. HMM Setup and Implementation

3.1 Model Architecture

Each activity (four in total) was trained with one Gaussian HMM, with four internal states for each, to represent the sub-phases of the activity (e.g., the phases in the gait cycle in walking). For a particular observation, classification chooses the model that has the maximum log-likelihood. Covariance reduction using diagonal was performed to increase training stability and reduce number of parameters.

The five components of the HMM are: The hidden states Z represent the activity; the observations X are 35-dimensional feature vectors; the transition matrix A specifies state-to-state probabilities; the emission probabilities B are represented by Gaussians; and the initial state probabilities π are uniform.

3.2 Baum-Welch Training

The parameters were learned by the algorithm of Baum-Welch. The E-step calculates forward and backward probabilities to obtain estimates of state occupancies. The M-step re-estimates the initial probabilities, the emission parameters and the transition matrix. Training is terminated if the change of log-likelihood is less than $1e-4$. All models converged within 200 iterations with log-likelihoods of +26,657 (still), +11,980 (standing), +7,324 (walking), and -3,810 (jumping).

3.3 Viterbi Classification

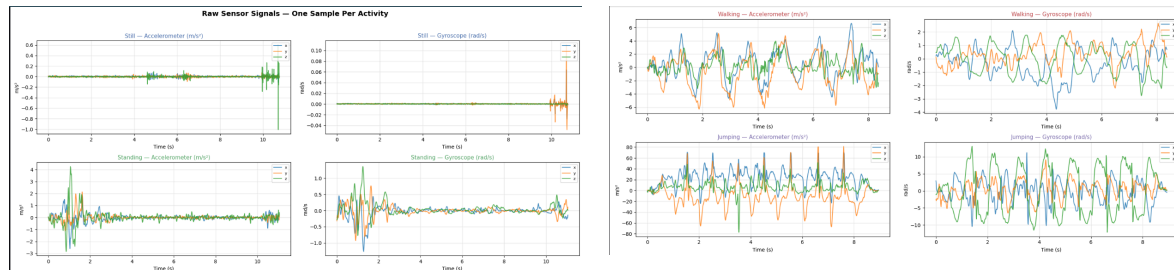
All HMMs were tested on each window and the best log-likelihood model was chosen for each window:

Training accuracy was 96.85%.

4. Results and Interpretation

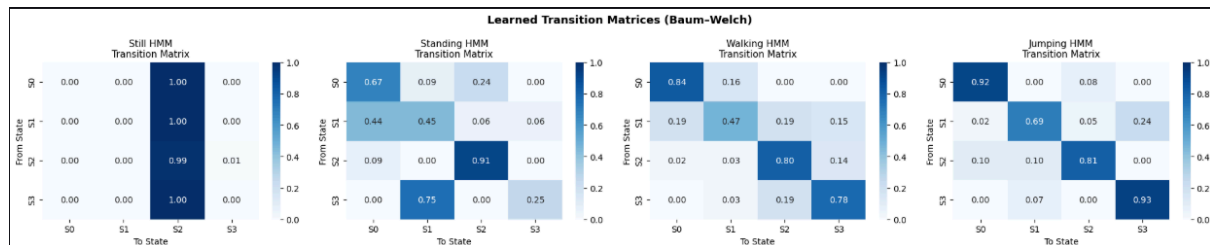
4.1 Raw Signal Analysis

There was no significant change in the amplitude of the near-flat signals during still produced signals, rhythmic oscillations at 1-2Hz occurred during walking, and large amplitude oscillations occurred during jumping due to landing impacts (Figure 1).



4.2 Transition Matrix Analysis

Still and standing are highly diagonal, signifying that they are stationary. Transitions between cyclic gait phases are distributed in a walker. The jumping exercise has the most evenly distributed transitions that represent quick shifts between the launch, airborne, and landing phases of the movement (Figure 2).

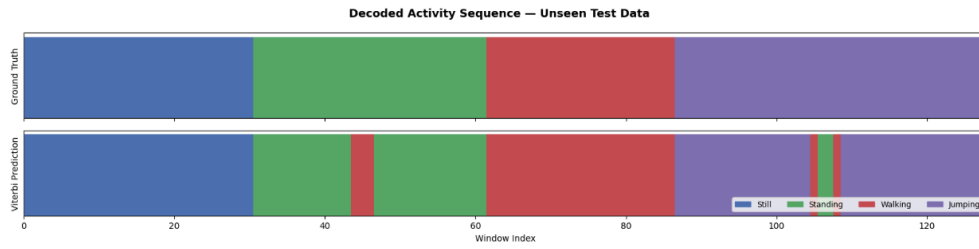


4.3 Evaluation on Unseen Test Data

The model was tested on 128 windows of 8 unseen recordings. The accuracy of the test set was 94.53%, with a decrease of just 2.32 points from the training set, showing good generalisation. Still earned all marks. The walking was the only test that had 100% sensitivity, and there was only a little standing confusion. The one that missed the most was standing (3 windows were classified as walking). Jumping was perfectly specific with 4 boundary windows being misclassified (Figure 3).

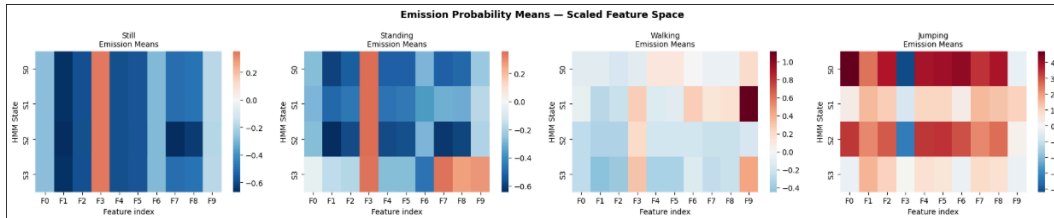
Activity	N Samples	Sensitivity	Specificity	Overall Accuracy
Still	31	1.000	1.000	1.000

Standing	31	0.903	0.979	0.961
Walking	25	1.000	0.951	0.961
Jumping	41	0.902	1.000	0.969
Overall	128	—	—	0.945



4.4 Emission Probability Analysis

The emission intensities of these are almost neutral in both still and standing. Walking and jumping displays great feature contrast, and hence, good differentiation of learned representations (Figure 4).



5. Discussion and Conclusion

The activity of “Still” was the simplest to classify due to its flat signal profile. Walking was reliably identified because there is a consistent rhythmic FFT signature. Standing and jumping were most challenging. Standing was often confused with walking at transitions where the natural swaying in the body generated oscillations similar to those of the first few steps of walking. Errors were made when the subject was moving in or out of the activity (jumping errors).

Fast transient events, such as jump landings were adequately recorded at 100 Hz. An even lower rate would risk aliasing these high frequency components. Windowed feature averaging was used to reduce the sensor noise. The small drop in accuracy from training to test may have been due to the variability in phone position from session to session.

Future work involves recording more data per activity, incorporating techniques like jerk and entropy to improve the standing/walking boundary, having more than one participant record the data for more generalisation, and using majority voting over a rolling prediction window to smooth boundary misclassifications.

The results of this project were successfully applied in HMM based activity recognition on a smartphone for physiotherapy rehabilitation monitoring. The pipeline of feature extraction, Z-score normalisation and per-activity Gaussian HMM training yielded 96.85% training accuracy and 94.53% test accuracy on unseen data. The small gap between the two performance and the physically interpretable errors verify the appropriateness of HMMs for sequential activity classification based on sensors.

References

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